

QUESTION PAPER

Name of the Examination: CAT – 2 (Winter 2021-2022)

Course Code: CSE3004

Course Title: Design and Analysis of Algorithms

Slot: D2+TD2

Date of Exam: 21.4.2022

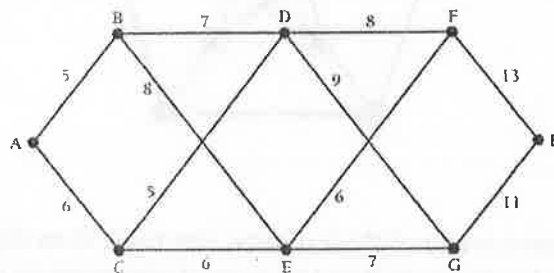
Duration: 90 min

Total Marks: 50

Class ID: AP2021225000243

1. The diagram below shows roads connecting towns A,B,C,D,E,F,G and H. The numbers on each arc represent the time, in minutes, required to travel along each road. Peter is delivering books from his base at A to H. Use Dijkstra's algorithm to find the minimum time for Peter's journey.

(10M)



2. A) Apply Warshall's algorithm to find the transitive closure of the digraph defined by the following adjacency matrix: (5M)

```

0 1 0 0
0 0 1 0
0 0 0 1
0 0 0 0

```

- B) Determine the cost and structure of optimal binary search tree for a set $K=(k_1, k_2, k_3)$ 3 keys. (5M)

P_i is the probability for searching values in K and Q_i is the probability for searching values not in K .

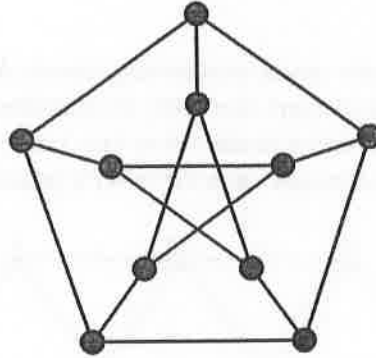
	0	1	2	3
P_i		1/7	1/7	1/7
Q_i	1/7	1/7	1/7	1/7

3. Assume that we have a knapsack with max weight capacity, $W = 16$. Our objective is to fill the knapsack with items such that the benefit (value or profit) is maximum. Solve using greedy methodology with algorithm? (5M)

ITEM	WEIGHT	VALUE
i1	6	6
i2	10	2
i3	3	1
i4	5	8
i5	1	3
i6	3	5

4. The Petersen graph is most drawn as a pentagon with a pentagram inside, with five spokes. Find the chromatic number (minimum value) of the following graph with state space tree.

(10M)



5. Given an integer array of N elements, the task is to divide this array into K non-empty subsets such that the sum of elements in every subset is same. All elements of this array should be part of exactly one partition. Solve the problem using backtracking mechanism for the given data set and draw state space tree? Array A= {5,7,10,12,15,18,20} and Sum=35 (10M)

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	Marks
Q1	3	3	PO1, PO2, PO3, PO4, PO9, PO10, PO11, PO12	10
Q2	3	3	PO1, PO2, PO3, PO4, PO9, PO10, PO11, PO12	10
Q3	3	3	PO1, PO2, PO3, PO4, PO9, PO10, PO11, PO12	10
Q4	4	4	PO1, PO2, PO3, PO4, PO5, PO11	10
Q5	4	4	PO1, PO2, PO3, PO4, PO5, PO11	10



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QUESTION PAPER

Name of the Examination: CAT (FALL 2022-2023)

Course Code: CSE 3004

Course Title: Design and Analysis of Algorithms

Slot: C1+TC1

Date of Exam: 02/11/2022

Duration: 90 min

Total Marks: 50

Answer ALL Questions

1. Solve and analyse the optimal binary search tree using dynamic programming approach with the following input values. (Write all steps properly)
Key= (5, 10, 15, 20) and the corresponding frequency i.e. Frequency= (4,2,6,3) (10 Marks)

2. For the following matrices find the order of parenthesization for the optimal matrix chain multiplication. (10 Marks)

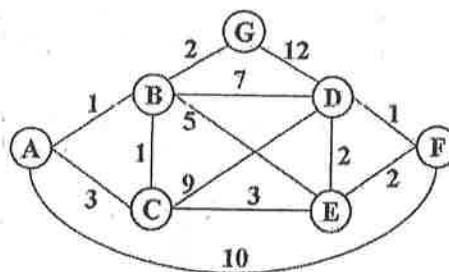
$$A_1 = 6 \times 5$$

$$A_2 = 5 \times 7$$

$$A_3 = 7 \times 10$$

$$A_4 = 10 \times 15$$

3. In Dijkstra algorithm why do we use a priority queue as opposed to a regular first in first out queue? Solve the given example on which Dijkstra algorithm fails to compute shortest paths if a first in first out queue is used. (10 Marks)



4. Given the following array [48, 51, 72, 34, 16, 83, 07, 67, 21, 24, 45, 10] and assume that quick sort will be used to sort this array in ascending order. Select the value for the last element of the array, such that the partitioning performed by the quick sort is most balanced. Calculate the worst-case complexity through recurrences relation. Also, explain why this makes quick sort perform efficiently. (10 Marks)

5. Why the recurrences relation required? Solve the following recurrence relation. (10 Marks)

- i. $T(n) = 2T(n/4) + (\text{root}(n))$
- ii. $T(n) = T(n-1) + \log n$

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	3	1,2,3			10
Q2	3	3	1,2,3			10
Q3	3	3	1,2,3			10
Q4	2	2	1,2			10
Q5	1	1	1,2			10

QUESTION PAPER

Name of the Examination: CAT (FALL 2022-2023)

Course Code: CSE3004

Course Title: Design and Analysis of Algorithms

Slot: D1+TD1

Date of Exam: 03/11/2022

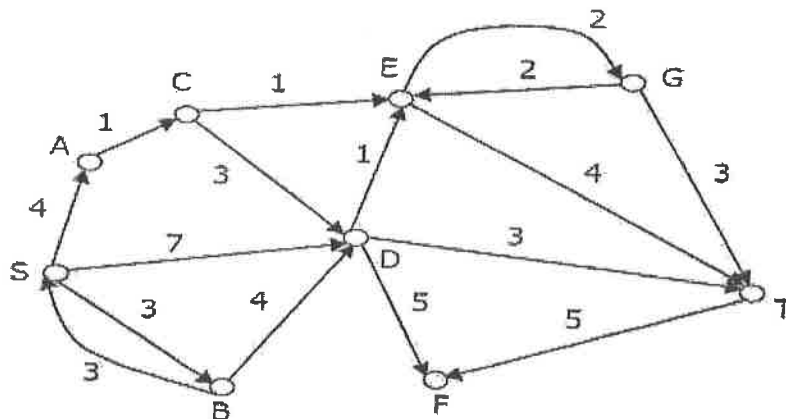
Duration: 90 min

Total Marks: 50

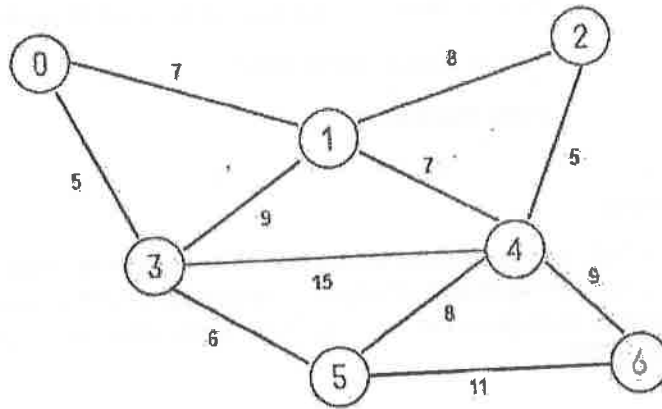
Instructions:

1. Answer all the questions.

- Q1.** (a) State Matrix Chain Multiplication Problem. Write Dynamic Programming Algorithm for Matrix Chain Multiplication Problem. Analyse how the time complexity of brute force chain matrix multiplication i.e. $O(2^n)$ has drops down to $O(n^3)$ using dynamic programming. (5M)
- (b) Using Dynamic Programming, find the optimal sequence for multiplying the chain of matrices $\langle A_1 A_2 A_3 \rangle$ whose dimensions are $\langle 8 \times 5 \rangle$, $\langle 5 \times 3 \rangle$, and $\langle 3 \times 6 \rangle$ respectively. (5M)
- Q2.** A thief robbing a store finds n items. The i th item is worth P_i and weight W_i positive integer. The thief must either take the item or leave it. He cannot take a fraction of the item. He can carry at least m kilos in his knapsack. Given, $n=4$, $m=5$, $W_i = \{2, 1, 3, 2\}$ and $P_i = \{12, 10, 20, 15\}$. Find the optimal solution for the given Knapsack problem. Also find the objects to be included in Knapsack in terms of 0 and 1. (10M)
- Q3.** (a) Let $T(n) = 4T(n/2) + n^2$. Solve the recurrence relation. (4M)
- (b) Find the shortest path from s to all other vertices and in the following graph using Dijkstra's Algorithm. (6M)



- Q4. (a) Consider a graph where the node represents houses and edge represents pipe lines. Our task is to set up lines in such way that all the houses are connected and the cost of setting the whole connection is minimum. The safe edge (pipe lines) added to a graph may connects two distinct components. Write an algorithm to solve the above problem. Analyse its complexity. (5M)
- (b) Compute the Minimum Spanning Tree and its cost for the following graph using Kruskal's Algorithm. Indicate each step clearly. (5M)



- Q5. Define optimal Binary Search Tree. Consider four keys A, B, C and D to be searched with probabilities 0.1, 0.2, 0.4, and 0.3 respectively. Find the cost for optimal binary search tree. Also draw the optimal binary search tree. (10M)

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	3	3	1,2,3			10
Q2	2	2	1,2,3			10
Q3	1,2	1,2	1,2,3			10
Q4	2	2	1,2			10
Q5	3	3	1,2			

DESIGN AND ANALYSIS OF ALGORITHMS

Name of the Examination: CAT

School	SCOPE		
Course Code	CSE3004	Course Title	Design and Analysis of Algorithms
Semester	FALL 2022-23	Date of exam	07/11/2022
Slot	G2+ TG2	Faculty Name & ID	
Duration	90 min	Total Marks	50

Q1. A) Find the order of the growth of the following functions use any asymptotic notation? (5M)

a) $\sum_{i=1}^n 2^i + i^2$ b) $\sum_{i=1}^n \sum_{j=i+1}^{n-1} 1$

B) Find the solution of the Job assignment problem to the following instance using brute force technique? (5M)

	Job1	Job2	Job3	Job4
Person1	6	4	3	7
Person2	9	2	7	8
Person3	1	8	5	8
Person4	7	6	9	4

Q2. Find the solution of the Knapsack problem to the following instance using brute force technique? (10M)

	Weight	Profit	Bag capacity m=16
Object 1	5	4	
Object 2	6	2	
Object 3	4	8	
Object 4	7	6	

Q3. Calculate average case time complexity of successful and unsuccessful search for the following keys in binary search algorithm using divide and conquer? Draw recursion tree of algorithm and calculate its depth? (10M)

{ -29,-22,-19,-15,-11,-10,1,3,4,5,7,8,11,14,16,18,20,21,22,27 }

Q4. Apply Quicksort to sort the list {U,N,I,V,E,R,S,I,T,Y} in alphabetic order using divide and conquer technique. Draw the tree of the recursive calls made with selection of pivot as first element? (10M)

Q5. A Salesman must visit from city to city to maintain his accounts. He has to leave his home city and visit other cities once and return home. The cost of going from city to city is shown in table. Find the least route? Use dynamic programming methodology to solve problem? (10M)

	City1	City2	City3	City4
city1	0	2	9	10
city2	1	0	6	4
city3	15	7	0	8
City4	6	3	12	0

QP Mapping

Q. No.	Module Number	CO Mapped	PO Mapped	Marks
Q1	1,2	CO1, CO2		10 Marks
Q2	2	CO2		10 Marks
Q3	2	CO2		10 Marks
Q4	2	CO2		10 Marks
Q5	3	CO3		10 Marks

QUESTION PAPER

Name of the Examination: FAT (Winter 2021-2022)

Course Code: CSE3004

Slot: D2+TD2

Course Title: Design and Analysis of Algorithms

Date of Exam: 26/05/2022

Duration: 120 min

Total Marks: 60

Q1. A Salesman must visit from city to city to maintain his accounts. He must leave his home city and visit other cities once and return home. The cost of going from city to city is shown in table. Find the least route? Use LC branch and bound methodology to solve problem?

(15M)

	City1	City2	City3	City4	City 5
city1		7	3	12	8
city2	3		6	14	9
city3	5	8		6	18
City4	9	3	5		11
City5	18	14	9	8	

Q2. Draw the portion of state space tree generated by FIFO branch and bound for the solution of following knapsack instance?

(15M)

$N=5, m=12$

	Profits	Weights
Object 1	10	4
Object 2	15	6
Object 3	6	3
Object 4	8	4
Object 5	4	2

Q3. The 15-puzzle consists of 15 numbered tiles on a square frame with a capacity of 16 tiles. We are given an initial arrangement of the tiles, and the objective is to transform this arrangement into the goal arrangement of through a series of legal moves. The only legal moves are ones in which a tile adjacent to the empty spot (ES) is moved to ES. Solve the problem of transformation from initial to goal arrangement (shown in below figure) using LC branch and bound with state space tree.

(10M)

Initial arrangement				Goal arrangement			
1	2	3	4	1	2	3	4
5	6		8	5	6	7	8
9	10	7	11	9	10	11	12
13	14	15	12	13	14	15	

Q4. Let there be N workers and N jobs. Any worker can be assigned to perform any job, incurring some cost that may vary depending on the work-job assignment. It is required to perform all jobs by assigning exactly one worker to each job and exactly one job to each agent in such a way that the total cost of the assignment is minimized. Solve the following instance using LC branch and bound with state space tree. (10M)

	J1	J2	J3	J4
W1	82	83	69	92
W2	77	37	49	92
W3	11	69	5	86
W4	8	9	98	23

Q5. Determine the cost and structure of optimal binary search tree for a set $K = (k_1, k_2, k_3)$ 3 keys. (10M)

P_i is the probability for searching values in K and Q_i is the probability for searching values not in K .

	0	1	2	3
P_i		1/7	1/7	1/7
Q_i	1/7	1/7	1/7	1/7

QP MAPPING

Q. No.	Module Number	CO Mapped	PO Mapped	Marks
Q1	6	CO6	PO3, PO4, PO5, PO6, PO9	15
Q2	6	CO6	PO3, PO4, PO5, PO6, PO9	15
Q3	6	CO6	PO3, PO4, PO5, PO6, PO9	10
Q4	6	CO6	PO3, PO4, PO5, PO6, PO9	10
Q5	3	CO3	PO2, PO3, PO4, PO5, PO6, PO11	10

QUESTION PAPER

Name of the Examination: FAT (Long Summer 2022)

Course Code: CSE3004

Course Title: Design and Analysis of Algorithm

Slot: A+TA+TAA

Date of Exam: 28-07-2022

Duration: 120 min

Total Marks: 60

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Note: All questions carry equal marks.

- Q.1. a) What are the basic asymptotic efficiency classes? Discuss Fundamentals of the analysis of algorithm efficiency elaborately. (6)
 b) What is algorithm visualization? (2)
 c) What is order of growth? (2)
- Q.2.a) Discuss important problem types that you face during Algorithm Analysis. (5)
 b) Explain in detail greedy techniques with an example. (5)
- Q.3. Outline the stable marriage problem with example. What is bipartite graph? Is the subset of bipartite graph is bipartite? Outline with example. (10)
- Q.4. Define sum of subset Problem. Explain Promising and Non-Promising Node. (10)
- Q.5. State the reason for terminating search path at the current node in branch and bound Algorithm. What are the searching techniques that are commonly used in Branch-and-Bound method. (10)
- Q.6. Give solution to Hamiltonian circuit using Backtracking technique. Elaborate how backtracking technique can be used to solve the n-queens problem. Explain with an example. (10)

QP Mapping

Q. No.	Module Number	COMapped	PO Mapped	PEO Mapped	PSO Mapped	Marks
Q1	1	CO5	1,2,3	1,2,3,4,9,10,11,12	1,2,3	10
Q2	2	CO5	1,2,3	1,2,3,4,9,10,11,12	1,2,3	10
Q3	4	CO6	1,2,3	1,2,3,4,9,10,11,12	1,2,3	10

Q4	5	CO6	1,2,3	1,2,3,4,9,10, 11,12	1,2,3	10
Q5	6	CO2	1,2,3	1,2,3,4,9,10, 11,12	1,2,3	10
Q6	5	CO2	1,2,3	1,2,3,4,9,10, 11,12	1,2,3	10



QUESTION PAPER

Name of the Examination: CAT (Summer 2022)

Course Code: CSE3004

Course Title: Design and Analysis of Algorithm

Slot: A+TA+TAA

Date of Exam: 01-07-2022

Duration: 90 min

Total Marks: 50

Instructions:

1. Assume data wherever necessary.
2. Any assumptions made should be clearly stated.

Note: All questions carry equal marks.

Q.1. a) What is the formula used in Euclid's algorithm for finding the greatest common divisor of two numbers? Write an algorithm to find the number of binary digits in the binary representation of a positive decimal integer. (6)

b) Mention the useful property, which can be applied to the asymptotic notations and its use? (2)

c) Compare the order of growth $n(n-1)/2$ and n^2 . (2)

Q.2.a) Write the Brute force algorithm to string matching. (5)

b) Write the algorithm for Iterative binary search. (5)

Q.3. Explain Merge Sort and quick sort with suitable example. Write down algorithms and compare them. (10)

Q.4. How Dynamic Programming is used to solve Knapsack Problem? Write down the algorithm. (10)

Q.5. a) Define the Single Source Shortest Path Problem. (5)

b) Solve the following graph for single source shortest path. (5)

